Secret-Learn — Privacy-Preserving ML Architecture

Applications, sklearn API, FL / SS / SL modes, JAX acceleration, secure devices, and infrastructure

Dashed frame = architecture zone
Solid frame = subsystem
Box = module and responsibility

PROJECT OVERVIEW

Secret-Learn exposes sklearn-compatible machine learning across federated, secret-sharing, and split-learning modes.
The architecture comprises the sklearn API, three privacy modes, JAX acceleration, SecretFlow devices, and infrastructure.

APPLICATION · multi-party use cases

- **Healthcare** — hospital collaboration, patient privacy
- **Finance** — cross-bank analysis, fraud detection
- **IoT / Edge** — edge intelligence, distributed learning
- **Research** — genomics, multi-institution
- **Government** — census analytics, cross-agency data

SKLEARN API · 191 algorithms

- **Linear Models** — 39 — Ridge · Lasso · SGD
- **Ensemble** — 18 — RF · XGBoost
- **Clustering** — 14 — KMeans · DBSCAN
- **Decomposition** — 14 — PCA · NMF · ICA
- **SVM** — 7 — SVC · SVR
- **Neural Networks** — 4 — MLP · BernoulliRBM
- **+ 95 More** — Neighbors · Trees, Naive Bayes

PRIVACY MODES · execution placement

FEDERATED LEARNING

- **Local PYU** — data stays local
- **HEU Aggregate** — secure averaging
- **FLLinearRegression** — federated_learning/
- **Horizontal FL** — 3–5×

SECRET SHARING

- **SPU MPC** — ciphertext compute
- **ABY3 / CHEETAH** — MPC protocols
- **SSPCA** — secret_sharing/
- **Maximum Privacy** — 1–2×

SPLIT LEARNING

- **Model Split** — across parties
- **HEU Activations** — protected tensors
- **SLMLPClassifier** — split_learning/
- **Vertical FL** — 2–4×

JAX ACCELERATION · xlearn bridge

- **JIT / XLA** — traced execution
- **jax.vmap** — batching · autodiff
- **xlearn Backend** — JAX-sklearn bridge
- **Smart Fallback** — original sklearn
- **CPU** — 1.5–2.5×
- **GPU** — 5–8×
- **TPU** — 9–15×
- **M1 / M2** — 2–4×

SECURE DEVICES · SecretFlow

SPU

- **MPC** — ABY3 · CHEETAH
- **Ciphertext** — zero plaintext leak

HEU

- **Schemes** — Paillier · CKKS
- **Aggregation** — gradient protection

TEE

- **Enclaves** — Intel SGX · AMD SEV
- **Occlum** — attestation

PYU

- **Party Local** — data ownership
- **JAX Local** — plaintext compute

INFRASTRUCTURE · runtime stack

- **SecretFlow** — Ray distributed, gRPC · DP-SGD
- **JAX / XLA** — CUDA / cuDNN, TPU runtime
- **sklearn** — NumPy · SciPy

FL, SS, and SL are three placements of the same sklearn API. JAX and SecretFlow devices PYU, HEU, SPU, and TEE form the compute boundary.